

VC-LFOs

In a music synthesizer, LFOs (Low Frequency Oscillators) serve to facilitate modulation of all kinds. It is they that truly imbue synthesizer sounds with their "vitality." The capability for voltage control of these modulation oscillators offers the user a significant expansion of their musical expressive possibilities (e.g., "densification" of phasing patterns, velocity-sensitivity of tremolo frequency, tracking of the vibrato with the pitch of the VCOs, etc.).

This LFO module provides two independent low-frequency modulation voltages (for FM, PWM, TM, CM, AM, etc.) and is, moreover, itself voltage-controllable. The concept of voltage control—introduced by Dr. Robert A. Moog—has been consistently applied here in a modified FORMANT LFO module (Figure 1), thereby opening up new possibilities for sound manipulation to the FORMANT player.

While optimal utilization of the PCB and front panel real estate allows for the accommodation of only two oscillators per board, this trade-off is acceptable given the provision for voltage control. Furthermore, three distinct waveforms are simultaneously available at the output of each oscillator, thereby enabling the realization of complex modulations.

The circuit board houses two independent LFOs, which function essentially as low-frequency function generators. At the output of LFO 1, square, triangle, and rising-edge sawtooth waveforms are available (the latter operating at double the frequency due to circuit design); at the output of LFO 2, in addition to the triangle wave, two different "sawtooths" are provided: one with a rising edge and one with a falling edge. For both LFOs, the frequency of the output signals is adjustable within a range of approximately 0.005 Hz to 20 Hz.

As can be seen in Figure 2, all signals range between -2.5 V and $+2.5\text{ V}$; an indicator LED makes the waveform of the triangle oscillation visible. This serves as a useful aid in configuring and assigning the respective modulation source—particularly in the case of very slow-moving LFOs.

Circuitry

The oscillator section for both LFOs is shown in Figure 3 (a and b). With the exception of a single resistor in each, the schematics are identical to those of the FONAMANT-1 LFOs featured in Part 9 of this series. However, the circuit's function is explained once again below: The actual oscillator consists of an integrator (IC1 or IC3) and an op-amp Schmitt trigger (A3 or A7). The oscillator's triangular waveform is generated via feedback from the Schmitt trigger's output to the integrator's input. The voltage divider formed by R10/P3/R11 (R32/P7/R33) scales down the output voltage from A3 (A7)—which alternates between approximately $+14\text{ V}$ and -14 V —to $+2.5\text{ V}$ and -2.5 V , respectively, at the wiper of P3 (P7). Resistors R8 and R9 (R30 and R31) form a voltage divider for the voltage present between the output of IC1 (IC3) and the wiper of

P3 (P7); consequently, exactly half of this voltage is always present at the non-inverting input of A3 (A7) ($R8 = R9 = R30 = R31 = 100\text{ k}\Omega$). Strictly speaking, op-amp A3 (A7) functions as a comparator: its output switches to approximately +14 V as soon as the non-inverting input becomes positive relative to the inverting input (which is held at ground), and to approximately -14 V as soon as the non-inverting input becomes negative. Due to the feedback of the output voltage—scaled down to +2.5 V—to the voltage divider R8/R9 (R30/R31), the output of A3 (A7) switches to approximately +14 V when the output voltage of IC1 (IC3) rises to +2.5 V (the upper switching threshold of the trigger), and to approximately -14 V when the integrator's output voltage reaches -2.5 V (the lower switching threshold).

Via R36 and P1 (R37 and P8), the comparator's output voltage is fed to the input of IC1 (IC3), thereby reversing the direction of the voltage change at the integrator's output whenever the switching thresholds are reached. A triangular waveform is generated at the integrator's output, the amplitude of which corresponds to the hysteresis of the Schmitt trigger. The integrator's input voltage determines the rise and fall rates of the output voltage waveform, while the frequency is adjustable via potentiometer P1 (P8). The square-wave voltage generated by the Schmitt trigger is buffered by the op-amp A4 to provide a low-impedance output. A simple converter circuit derives a sawtooth waveform from the triangular waveform. As long as the voltage at the wiper of P3 (P7) is negative (specifically, during the rising edge of the triangle wave), T1 (T2) is cut off; OpAmp A2 (A6) then operates as a non-inverting amplifier with a gain of 1. During this time, its output voltage is identical to the triangle oscillation. With the positive edge of the square wave signal (at the moment the triangle oscillation reaches the upper switching threshold of the Schmitt trigger—see Figure 2), T1 (T2) becomes conductive and switches A2 (A6) into inverting mode, resulting in a negative edge (the falling slope of the sawtooth wave) at its output. Since A2 (A6) is now inverting, the subsequent falling slope of the triangle oscillation produces a mirror-image rising slope at the sawtooth output. Figure 2 illustrates the voltage waveforms and the relative phase relationships of the LFO output signals. The LED display—comprising D2 and D4—which has already proven its worth in several FORMANT modules and serves as more than just a visual gimmick, provides a "linearized" visual indication.

The only difference between the two LFOs lies in the utilization of one of the four OpAmps within IC2 or IC4, respectively. While LFO 1 employs OpAmp A4 to provide low-impedance buffering for the square wave output, LFO 2—which dispenses with a dedicated square wave output—instead utilizes the "spare" amplifier, A8, to invert the sawtooth signal "around the zero line," thereby producing a sawtooth wave with a falling slope at its output.

Figure 4 depicts the control voltage section, which is required in duplicate. Parameter P9 allows for the adjustment of the overall tuning of both LFOs, while P10 and P10' serve to attenuate the external control voltage signal. P9 must be a stereo potentiometer. The control voltage is fed into the feedback loop of A3 (A7) and IC1 (IC3). Resistor R36 (R37) isolates the output of comparator A3 (A7) from the control voltage. Since the input voltage of integrator IC1 (IC3) determines the rise and fall rates at its output—and thus the frequency of the LFO—this frequency now also depends proportionally on the magnitude of the control voltage. It is also

worth noting that the LFOs continue to "oscillate" even when no external control voltage is applied.

Construction and Calibration

Due to the rather compact layout of the LFOs on the circuit board, care should be taken when soldering in the components. The board layout and component placement guide are shown in Figure 5. For C1 and C2, it is absolutely essential to use low-leakage film capacitors (Siemens MKH, Wima MKS). For space reasons, rotary potentiometers with 4 mm shafts should be selected. Figure 6 presents a suggested front panel layout.

Each LFO features four calibration points: signal amplitude adjustment (P3 or P7), integrator offset adjustment (P2 or P5), setting of R16 or R17 to define the maximum period duration, and adjustment of the LED indicator (P4 and P6).

The calibration procedure is identical—in both method and sequence—for both LFOs.

Signal Amplitude Adjustment

Before calibration, set trimmer P2 (or P5) to its center position and use P1 (or P8) to set the maximum frequency (wiper at R36 or R37). Next, measure the triangle wave output voltage using an oscilloscope and adjust it to +2.5 V. Subsequently, check the amplitude and waveform of the other outputs and, if necessary, readjust P3 (or P7).

Integrator Offset Adjustment

As a general rule, trimmer P2 (P5) can be left in its center position. However, if the oscillators "prematurely" cut out at low frequencies, a precise offset adjustment is necessary: Connect the wiper terminal of P1 (P8) and R1 (R19) to ground. This effectively connects both inputs of IC1 (IC3) to ground. Connect a multimeter to output Pin 6 (of IC1 or IC3) and select a DC voltage measurement range of 30 V or 50 V. The voltage at the triangle output will now lie between +15 V and –15 V and will drift in either a positive or negative direction. Should the voltage "get stuck" at one of these two maximum values (+15 V), discharge capacitor C1 (C2) through a 1 k Ω resistor. Now, adjust P2 such that the voltage change (ideally near 0 V) comes to a standstill. Next, select a lower measurement range, discharge C1 (C2) again, and fine-tune the setting of P2 to minimize any voltage change. Repeat this process until, even on the lowest DC measurement range (e.g., 100 mV), the voltage at the triangle output fluctuates by only a few mV around the zero point.

Calibration of R16 (R17)

If the maximum period duration (even after performing the offset adjustment) is less than 3 minutes, or is otherwise unsatisfactory, it can be increased by reducing the value of R16 (R17) may be increased. However, the resistance value should not fall below 10 Ω .

LED Display Calibration

Trimmer P4 (P6) should be adjusted such that the LED follows the contour of the triangle wave as "linearly" as possible—i.e., ensuring that neither bright nor dark gaps occur.

Applications

The two independent low-frequency control voltage sources each generate three waveforms, all of which are simultaneously available. They are particularly well-suited for all types of modulation: Frequency Modulation (FM; "Vibrato"), Pulse Width Modulation (PWM; "Phasing"), Timbre Modulation, Amplitude Modulation (AM; "Tremolo"), Clipping Modulation (CM), etc. By cross-modulating the control voltages of the individual LFOs, powerful—and in some cases musically unconventional—modulation patterns can be created.

If a CV/Gate patchbay is available, the CV signal can be utilized as a control voltage for the VC-LFOs. When playing, for example, PWM-modulated square waves, this ensures a slower modulation rate for lower notes and a correspondingly faster rate for higher notes—a result that sounds significantly more pleasing (and is perceived as more "natural") than a fixed modulation frequency. Such automatic modulation control is particularly interesting when using sequencers. Furthermore, the VC-LFOs can, of course, be used for any of the configurations involving LFOs described in *FORMANT Book 1*.

References:

- M. Bertuch: VCLFO, *Patchcord*, No. 3, p. 16
- C. Chapman: *FORMANT-Musiksynthesizer*, Elektor Verlag
- R.A. Moog: Voltage Controlled Electronic Music Modules, *JAES*, Vol. 13, No. 3, pp. 200–206 (July 1965)

VC-LFOs Bill of Materials

Resistors (Carbon Film, 5%):

- R1, R2, R19, R20 = 68 k Ω
- R3, R4, R6, R8, R9, R21, R22, R24, R30, R31, R38, R38', R40, R40', R41, R41' , R42, R42' = 100 k Ω
- R5, R23 = 47 k Ω
- R7, R12, R13, R28, R29, R34 = 1 k Ω
- R10, R32 = 3 k Ω
- R11, R15, R33, R35 = 470 Ω
- R14, R18, R39, R39' = 22 k Ω
- R16, R17 = 47 Ω (Reference value; see text)
- R25, R27 = 10 k Ω
- R26 = 4.7 k Ω

- R36, R37 = 10 k Ω

Potentiometers:

- P1, P8 = 100 k Ω (Logarithmic) (4 mm shaft)
- P2, P4, P5, P6 = 10 k Ω (Trimmer)
- P3, P7 = 1 k Ω (Trimmer)
- P9 = 100 k Ω (Linear stereo version, 4 mm shaft)
- P10, P10' = 100 k Ω (Linear, 4 mm shaft)

Capacitors:

- C1, C2 = 1 μ F (MKM, MKS)
- C3, C4 = 10 μ F / 25 V

Semiconductors:

- IC1, IC3 = μ A 741C (Mini-DIP)
- IC2, IC4, IC5 = LM 324N
- T1, T2 = BC 107C, BC 547C, or similar
- D1, D3, D5, D5', D6, D6' = DUS
- D2, D4 = LED

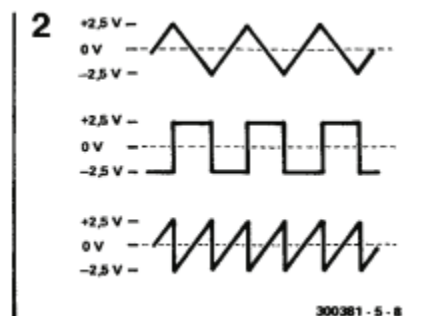
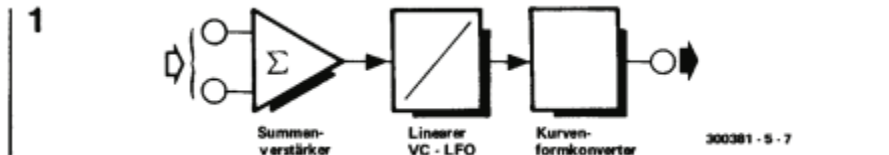
Miscellaneous:

- 8 x 3.5 mm Jack Sockets
- 5 x Rotary Knobs (4 mm)
- 1 x 31-pin Header Strip or Solder Pins

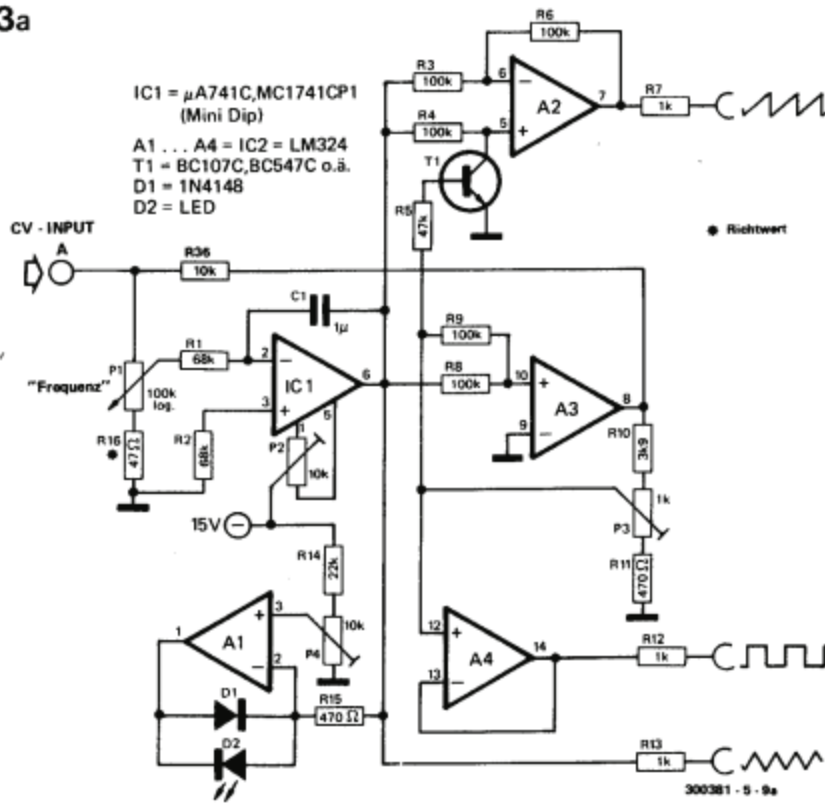
Figures

- Figure 1. Schematic structure of a voltage-controlled LFO based on the Moog modular concept.
- Figure 2. Waveforms of the LFO output signals. Due to the design of the triangle/sawtooth converter, the sawtooth wave exhibits twice the frequency of the triangle and square waves; however, this does not result in any musically disruptive effects.
- Figure 3. Schematic diagram of the LFO module, which contains two independent modulation oscillators. Figure 3a shows one oscillator generating triangle, square, and sawtooth waves. The second modulator circuit is shown in Figure 3b; in addition to the triangle voltage, it provides sawtooth signals with both rising and falling slopes.
- Figure 4. Circuit diagram of the control voltage section for the VC-LFOs. This circuit is duplicated on the printed circuit board (Figure 5).

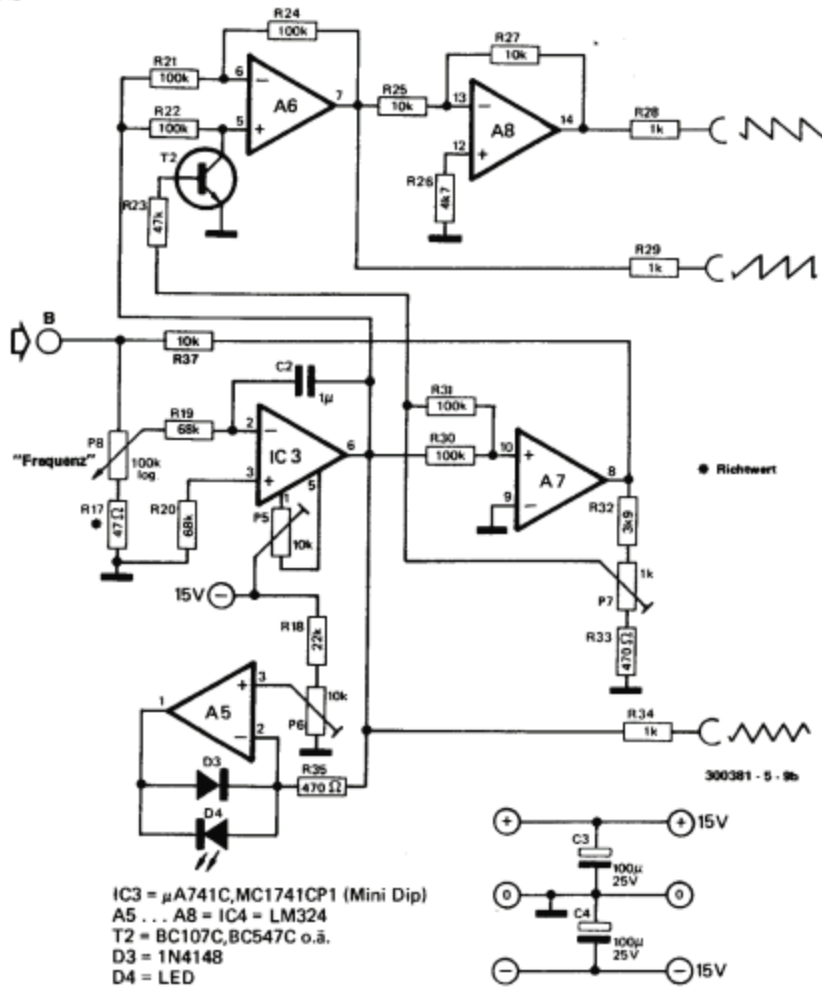
- Figure 5. Layout and component placement guide for the VC-LFO module. The following point must be observed when populating the PCB: Resistor R37 is soldered to the solder-side (trace-side) of the board.
- Figure 6. Suggested front panel design for the VC-LFO module.
- Figure 7. Two simple basic wiring configurations for the VC-LFO module.
- Figure 7a illustrates how the modulation frequency tracks the "pitch" of the keys. The wiring configuration shown in Figure 7b "compresses" the modulation pattern of the VC-LFOs.



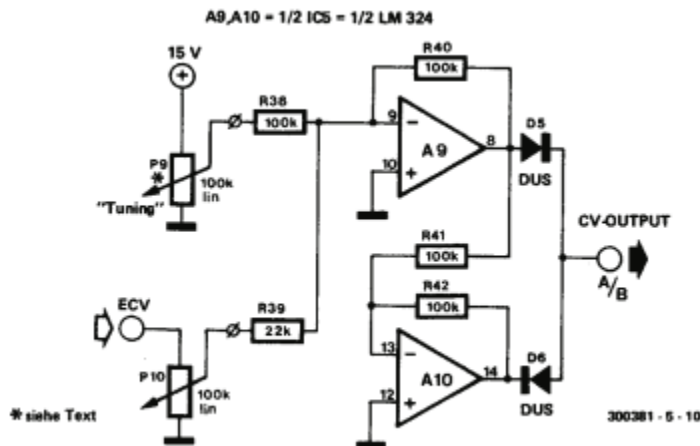
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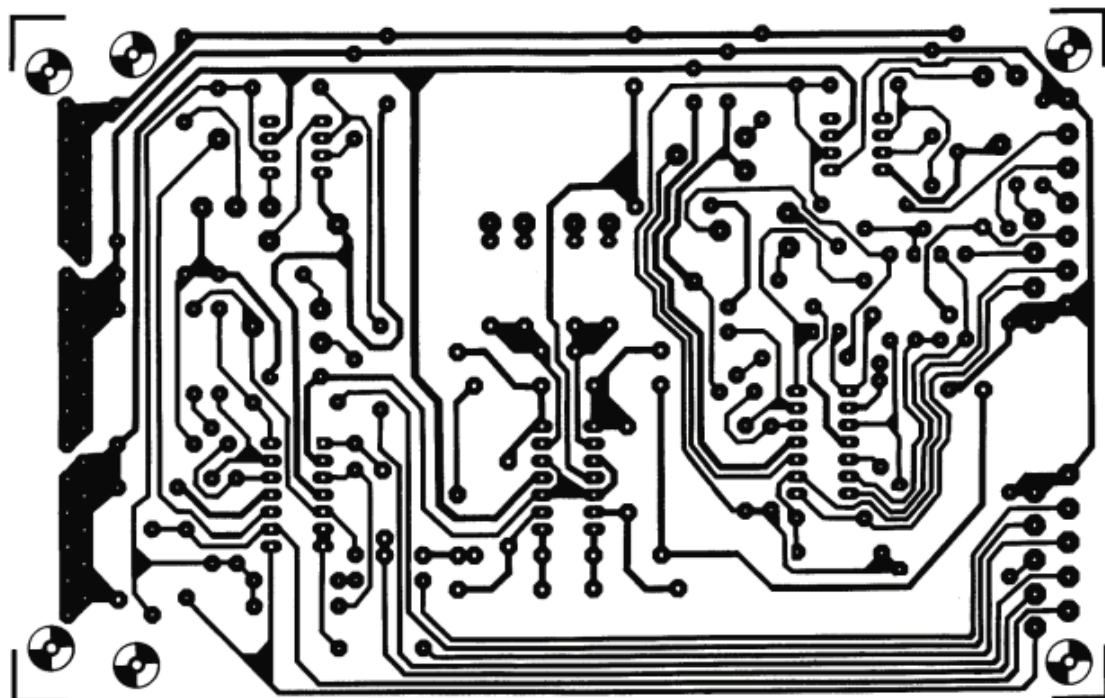
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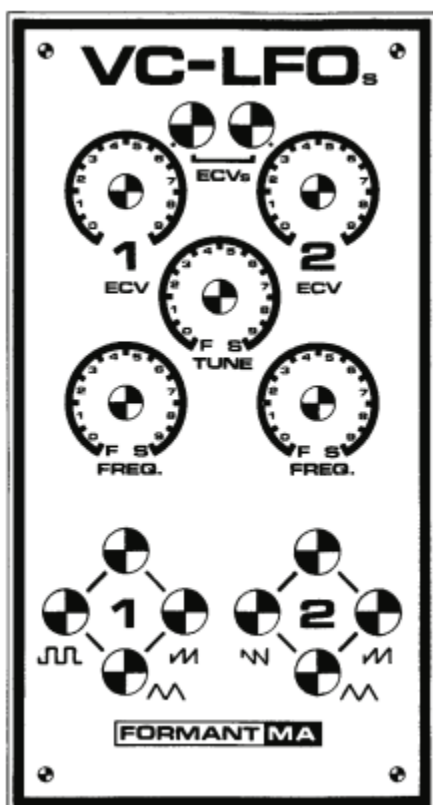
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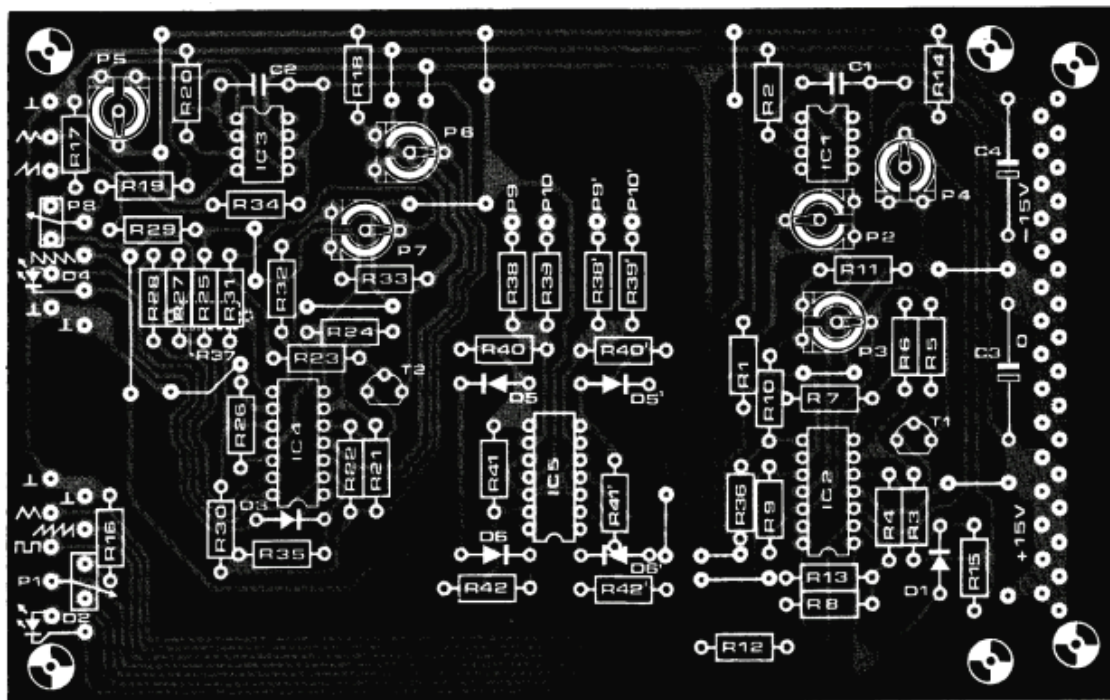


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7a

